

ASSIGNMENT CODE

```
df <- read_csv("data_for_analysis.csv") %>%
  select(-any_of("record_id"))

outcome_col <- "outcome"

# ----- FUNCTIONS -----

check_normal <- function(x) {
  x <- na.omit(x)
  if(length(x) < 3) return(FALSE)
  shapiro.test(sample(x, min(500, length(x))))$p.value > 0.05
}

make_summary <- function(x) {
  if (check_normal(x)) {
    paste0(round(mean(x, na.rm=TRUE),2), " ± ", round(sd(x, na.rm=TRUE),2))
  } else {
    paste0(round(median(x, na.rm=TRUE),2),
           " (",
           round(quantile(x,0.25,na.rm=TRUE),2), "-",
           round(quantile(x,0.75,na.rm=TRUE),2), ")")
  }
}

# =====
```

```

# Q1–Q2 ( NOT MAKING USE OF LIPIDS5)

# =====

df1 <- df %>% select(-any_of("lipids5"))

num_vars <- df1 %>% select(where(is.numeric)) %>% select(-outcome) %>% names()

# ---- Q1: Distribution ----

dist_results <- data.frame()

for (v in num_vars) {
  for (g in unique(df1[[outcome_col]])) {
    x <- df1 %>% filter(.data[[outcome_col]] == g) %>% pull(v)

    dist_results <- rbind(dist_results,
      data.frame(
        Variable = v,
        Outcome = g,
        Distribution = ifelse(check_normal(x), "Normal", "Non-normal")
      ))
  }
}

print(dist_results)

write.csv(dist_results, "Distribution_table_1.csv", row.names = FALSE)

```

```
# ---- Q2: Descriptive table ----
```

```
table_list <- list()
```

```
for (v in num_vars) {
```

```
  tmp <- df1 %>%
```

```
    group_by(outcome) %>%
```

```
    summarise(Value = make_summary(.data[[v]]), .groups="drop") %>%
```

```
    mutate(Variable = v)
```

```
  table_list[[v]] <- tmp
```

```
}
```

```
final_table <- bind_rows(table_list) %>%
```

```
  pivot_wider(names_from = outcome, values_from = Value)
```

```
print(final_table)
```

```
View(final_table)
```

```
write_csv(final_table, "Descriptive_table_1.csv")
```

```
# =====
```

```
# FOR EXTRA POINT Q1–Q3 with fixed lipids)
```

```
# =====
```

```
df2 <- df
```

```
df2$lipids5[is.na(df2$lipids5)] <- median(df2$lipids5, na.rm=TRUE)
```

```
num_vars2 <- df2 %>% select(where(is.numeric)) %>% select(-outcome) %>% names()
```

```
# ---- Extra Q1 ----
```

```
dist_results2 <- data.frame()
```

```
for (v in num_vars2) {
```

```
  for (g in unique(df2[[outcome_col]])) {
```

```
    x <- df2 %>% filter(.data[[outcome_col]] == g) %>% pull(v)
```

```
    dist_results2 <- rbind(dist_results2,
```

```
      data.frame(
```

```
        Variable = v,
```

```
        Outcome = g,
```

```
        Distribution = ifelse(check_normal(x), "Normal", "Non-normal")
```

```
      ))
```

```
    }
```

```
  }
```

```
print(dist_results2)
```

```
write.csv(dist_results2, "Extra_Distribution_table_2.csv", row.names = FALSE)
```

```
# ---- Extra Q2-Q3 ----
```

```
table_list2 <- list()
```

```
for (v in num_vars2) {  
  tmp <- df2 %>%  
    group_by(outcome) %>%  
    summarise(Value = make_summary(.data[[v]]), .groups="drop") %>%  
    mutate(Variable = v)  
  
  table_list2[[v]] <- tmp  
}
```

```
final_table2 <- bind_rows(table_list2) %>%  
  pivot_wider(names_from = outcome, values_from = Value)
```

```
print(final_table2)
```

```
View(final_table2)
```

```
write_csv(final_table2, "Extra_Descriptive_table _2.csv")
```